Chapter 2

Mass Spectrometry Principles and Instrument Physics

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Abstract

Mass spectrometers operate as physical sensors that convert gas-phase ions into digitized mass-to-charge ratios (m/z) and ion-current intensities. Neutral molecules cannot be detected directly; they must acquire electric charge through electron ionization (EI), electrospray ionization (ESI), or matrix-assisted laser desorption/ionization (MALDI). Because the instrument measures m/z rather than uncharged mass, molecular identity must be derived by deconvoluting integer charge states (z) from isotopic peak spacing. Instrument mass analyzers---quadrupoles, time-of-flight (TOF), and Orbitraps---differ by orders of magnitude in resolving power and mass accuracy, setting physical boundaries on isobaric separation. Furthermore, the choice of acquisition strategy governs data completeness: data-dependent acquisition (DDA) introduces stochastic missingness through top-N precursor selection, whereas data-independent acquisition (DIA) captures multiplexed fragmentation arrays at the expense of computational deconvolution. This chapter presents the physical governing equations, charge deconvolution algorithms, resolving power thresholds, and matrix suppression kinetics that define mass spectrometry data structures in R 4.6.1.

Keywords: Mass-to-charge ratio, Charge state deconvolution, Resolving power, Isobaric separation, DDA vs DIA, Ion suppression, Stable isotope standards.

Chapter Overview and Architectural Roadmap

Table 2.1 outlines the physical principles and analytical consequences of mass spectrometry instrumentation.

Methodological Note

Tutor Warning: Mass-to-Charge is Not Molecular Mass

A mass spectrometer measures the dimensionless ratio m/z, not molecular mass. A single peak observed at $m/z = 450.25$ represents a 449.24 Da singly-charged ion ($z = 1$), an 898.49 Da doubly-charged peptide ($z = 2$), or a 1,347.73 Da triply-charged species ($z = 3$). Attempting database matching

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Table 2.1: Physical principles and data characteristics of mass spectrometry systems.

Physical Domain	Instrument Mechanism	Mathematical Formulation	Data Structure Impact
1. Ionization	Gas-phase charging (ESI/MALDI)	$m/z = (M + z \cdot m_{\text{proton}})/z$	Multi-charge envelopes
2. Analyzer	Trajectory dispersion in fields	$R = m/\Delta m_{\text{FWHM}}$	Peak width and ppm error
3. Acquisition	DDA stochastic precursor selection	Top-N instantaneous intensity rank	Stochastic missingness
4. Multiplexing	DIA windowed fragmentation	Swath stepping $\Delta m = 20$--25 Da	Chimeric fragment arrays
5. Matrix Effect	Droplet evaporation competition	$I_{\text{obs}} = S(t) \cdot k \cdot C_{\text{analyte}}$	Ion suppression loss

without resolving the charge state from carbon-13 isotopic spacing ($\Delta(m/z) \approx 1.00335/z$) yields false identifications.

2.1 The Physical Measurement: Mass-to-Charge Ratio

Electric and magnetic fields manipulate charged particles according to the Lorentz force:

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \tag{2.1}$$

Because acceleration $\mathbf{a} = \mathbf{F}/m$ depends directly on the ratio of charge to inertial mass (q/m), mass spectrometers separate ions by their mass-to-charge ratio (m/z).

2.1.1 Charge State Determination from Carbon-13 Isotope Spacing

In organic molecules, naturally occurring carbon contains approximately 98.93% ^{12}C and 1.07% ^{13}C. The mass difference between a ^{13}C atom and a ^{12}C atom is:

$$\Delta m_{^{13}\text{C}-^{12}\text{C}} = 1.003355 \text{ Da} \tag{2.2}$$

When an ion acquires integer charge z, the observed difference in m/z between the monoisotopic peak (M_0) and the first isotopic peak (M_1) shrinks in inverse proportion to z:

$$\Delta(m/z) = \frac{1.003355}{z} \implies z = \text{round}\left(\frac{1.003355}{\Delta(m/z)}\right) \tag{2.3}$$

Once the charge state z is inferred, the neutral monoisotopic mass M_{neutral} is reconstructed by subtracting the mass of the charge-carrying protons ($m_{\text{proton}} = 1.007276$ Da):

$$M_{\text{neutral}} = z \cdot (m/z)_{\text{obs}} - z \cdot m_{\text{proton}} \tag{2.4}$$

2.2 Mass Resolving Power and Mass Accuracy

Two orthogonal metrics define mass analyzer performance: resolving power (R) and mass measurement accuracy (δ_{ppm}).

2.2.1 Resolving Power Definition

The mass resolving power R describes the ability of an analyzer to separate adjacent spectral peaks of equal intensity. Under the Full Width at Half Maximum (FWHM) convention:

$$R = \frac{m}{\Delta m_{\text{FWHM}}} \implies \Delta m_{\text{FWHM}} = \frac{m}{R} \tag{2.5}$$

To separate two isobaric compounds with mass difference $\Delta m = |m_2 - m_1|$, the instrument resolving power must satisfy:

$$R_{\text{required}} \geq \frac{m_1}{\Delta m} \tag{2.6}$$

2.2.2 Mass Accuracy in Parts-Per-Million

Mass measurement error represents the deviation between the observed experimental peak apex and the theoretical elemental mass:

$$\delta_{\text{ppm}} = \frac{m_{\text{obs}} - m_{\text{theor}}}{m_{\text{theor}}} \times 10^6 \tag{2.7}$$

High-resolution analyzers operate at tolerances $|\delta_{\text{ppm}}| \leq 5.0$ ppm, enabling narrow extracted ion chromatogram (EIC) windows that exclude background chemical noise.

2.3 Acquisition Architectures: DDA versus DIA

Untargeted mass spectrometry operates under two distinct tandem acquisition regimes:

2.3.1 Data-Dependent Acquisition (DDA)

In DDA, each duty cycle begins with an MS1 survey scan. The instrument firmware identifies the N most abundant precursor peaks (Top-N protocol), isolates each ion sequentially using a narrow quadrupole transmission window ($\approx$ 1.0--1.5 Da), fragments it via collision-induced dissociation (CID) or higher-energy collisional dissociation (HCD), and records clean, single-precursor fragment spectra.

Analytical Implication: Because complex proteomes contain hundreds of co-eluting peptides per second, Top-N selection is stochastic. Lower-abundance precursors hover near the detection threshold; Poisson ion-arrival statistics determine whether a peptide is fragmented in a given run, creating stochastic missingness across cohort runs.

2.3.2 Data-Independent Acquisition (DIA)

In DIA, the instrument disregards instantaneous precursor intensities. Instead, it systematically steps through a series of wide precursor isolation windows (Δm = 20--25 Da) spanning the entire mass range. All ions within each window are fragmented simultaneously.

Analytical Implication: DIA provides 100% acquisition completeness across all runs. However, the resulting MS2 spectra are multiplexed mixtures of fragments from dozens of co-isolated precursors. Identification requires matched spectral libraries and algorithmic deconvolution (e.g., DIA-NN, Spectronaut).

2.4 Matrix Effects and Ion Suppression

In electrospray ionization (ESI), charged droplets formed at atmospheric pressure undergo solvent evaporation and Coulomb fission. Co-eluting non-volatile salts, phospholipids, or polymer contaminants compete with target analytes for droplet surface access and available charge.

This competition attenuates analyte ionization by an empirical suppression factor $S(t) \in [0,1]$:

$$I_{\text{obs}}(t) = S(t) \cdot k \cdot C_{\text{analyte}} \tag{2.8}$$

Severe suppression causes signals to drop below the limit of detection, generating Missing Not At Random (MNAR) patterns.

Correction via Stable Isotope Standards: Spiking a stable isotope-labeled (SIL) internal standard corrects this distortion. Because the SIL analog co-elutes with the native analyte, both experience the identical suppression factor $S(t)$:

$$\frac{I_{\text{analyte}}}{I_{\text{SIL}}} = \frac{S(t) \cdot k \cdot C_{\text{analyte}}}{S(t) \cdot k \cdot C_{\text{SIL}}} = \frac{C_{\text{analyte}}}{C_{\text{SIL}}} \tag{2.9}$$

The matrix suppression factor cancels algebraically, preserving accurate linear quantification.

2.5 Laboratory Workflow and Executable Implementations

The laboratory exercises demonstrate charge deconvolution from isotope clusters, resolving power modeling across mass analyzers, DDA stochastic missingness simulation, and ESI ion suppression kinetics in R 4.6.1.

2.5.1 Listing 2.1: Resolving Charge States from Isotope Spacing

R Lab Listing 1: Listing 2.1: Deconvoluting Charge States from Isotopic Deltas [R 4.4 / Bioconductor]

```
library(tibble)
library(dplyr)
library(tidyr)

m_proton <- 1.007276466879
c13_diff <- 1.003354835

# Simulated isotopic clusters: z = 1, 2, 3, 20
test_charges <- c(1L, 2L, 3L, 20L)
mz_monos     <- c(450.2500, 450.2500, 450.2500, 848.5824)

deconv_list <- list()
for (i in seq_along(test_charges)) {
  z_true   <- test_charges[i]
  mz_0     <- mz_monos[i]
  delta_mz <- c13_diff / z_true
  mz_1     <- mz_0 + delta_mz

  meas_delta    <- round(mz_1 - mz_0, 4)
  inferred_z    <- round(c13_diff / meas_delta)
  neutral_mass  <- inferred_z * mz_0 - inferred_z * m_proton

  deconv_list[[i]] <- tibble(
    True_Z       = z_true,
    Observed_MZ0 = mz_0,
```

```
    Delta_MZ     = meas_delta,
    Inferred_Z   = as.integer(inferred_z),
    Neutral_Mass = round(neutral_mass, 2)
  )
}

print(as.data.frame(bind_rows(deconv_list)))
```

```
True_Z Observed_MZ0 Delta_MZ Inferred_Z Neutral_Mass 1 1 450.2500 1.0034 1 449.24 2 2 450.2500 0.5017 2 898.49 3
3 450.2500 0.3345 3 1347.73 4 20 848.5824 0.0502 20 16951.50
```

2.5.2 Listing 2.2: Resolving Power and Isobaric Separation

R Lab Listing 2: Listing 2.2: Calculating Peak FWHM and Isobaric Thresholds [R 4.4 / Bioconductor]

```
analyzers <- tibble(
  Analyzer  = c("Unit Quadrupole", "Time-of-Flight", "Orbitrap Standard", "Orbitrap Ultra"),
  Nominal_R = c(1000, 30000, 60000, 240000)
) %>%
  mutate(FWHM_at_500 = round(500.0 / Nominal_R, 5))

print(as.data.frame(analyzers))

# Isobaric doublet: Glutamine (146.0691) vs Lysine (146.1055)
delta_iso <- abs(146.1055 - 146.0691)
req_R     <- round(146.0691 / delta_iso)
cat(sprintf("\nGlutamine vs Lysine Delta: %.4f Da | Required R >= %d\n", delta_iso, req_R))
```

```
Analyzer Nominal_R FWHM_at_500 1 Unit Quadrupole 1000 0.50000 2 Time-of-Flight 30000 0.01667 3 Orbitrap Standard
60000 0.00833 4 Orbitrap Ultra 240000 0.00208
Glutamine vs Lysine Delta: 0.0364 Da | Required R >= 4013
```

2.5.3 Listing 2.3: Simulating Stochastic Sampling in DDA

R Lab Listing 3: Listing 2.3: Quantifying Missingness Driven by TopN Selection [R 4.4 / Bioconductor]

```
set.seed(42)
n_prec <- 300; n_reps <- 5
true_abund <- rlnorm(n_prec, meanlog = 12, sdlog = 2.0)

dda_mat <- matrix(FALSE, nrow = n_prec, ncol = n_reps)
for (rep_i in 1:n_reps) {
  obs_int <- true_abund * rlnorm(n_prec, 0, 0.15)
  dda_mat[order(obs_int, decreasing = TRUE)[1:25], rep_i] <- TRUE
}

cat(sprintf("Precursors Detected in All 5 Replicates: %.1f%%\n",
            mean(rowSums(dda_mat) == n_reps) * 100))
cat(sprintf("Precursors Completely Missing (0 of 5):   %.1f%%\n",
            mean(rowSums(dda_mat) == 0) * 100))
```

```
Precursors Detected in All 5 Replicates: 5.3Precursors Completely Missing (0 of 5): 89.7
```

2.5.4 Listing 2.4: ESI Ion Suppression Kinetics

R Lab Listing 4: Listing 2.4: Calibration Curve Slope Attenuation Under Matrix [R 4.4 / Bioconductor]

```
conc <- c(1, 5, 10, 25, 50, 100)
resp_clean  <- 10000 * conc + rnorm(length(conc), 0, 5000)
resp_matrix <- 10000 * 0.35 * conc + rnorm(length(conc), 0, 3000)

fit_c <- lm(resp_clean ~ conc)
fit_m <- lm(resp_matrix ~ conc)

cat(sprintf("Clean Slope:  %.0f counts/nM (R2 = %.4f)\n", coef(fit_c)[2], summary(fit_c)$r.squared))
cat(sprintf("Matrix Slope: %.0f counts/nM (R2 = %.4f)\n", coef(fit_m)[2], summary(fit_m)$r.squared))
cat(sprintf("Signal Loss:  %.1f%%\n", (1 - coef(fit_m)[2] / coef(fit_c)[2]) * 100))
```

```
Clean Slope: 10012 counts/nM (R2 = 0.9999) Matrix Slope: 3461 counts/nM (R2 = 0.9996) Signal Loss: 65.4
```

2.6 Five Deadly Sins of Mass Spectrometry Data Interpretation

1. **Confusing m/z with Molecular Mass:** Assuming observed peaks represent singly-charged ions misidentifies multiply-charged peptides and proteins.

2. **Assuming DDA Completeness:** Treating missing values in DDA as technical errors rather than stochastic precursor sampling distorts differential abundance statistics.

3. **Overlooking Matrix Suppression:** Failing to monitor internal standards in complex matrices attributes technical ionization losses to biological downregulation.

4. **Neglecting Isobaric Interferences:** Using broad extraction windows on low-resolution analyzers merges distinct chemical species into single distorted peaks.

5. **Ignoring In-Source Fragmentation:** Treating source-generated fragments as biological metabolites inflates feature tables with artificial molecular entities.

2.7 Pre-Flight Diagnostic Checklist

Table 2.2 provides an audit checklist to verify instrument parameters prior to data interpretation.

Table 2.2: Diagnostic checklist for mass spectrometry instrument configurations.

Category	Diagnostic Check	Acceptance Criteria	Status
Charge	Isotope cluster spacing verified.	Peak spacing $\Delta(m/z) \approx 1.00335/z$.	[ ]
Resolution	Resolving power adequate for target isobars.	$R \geq m/\Delta m$ for critical metabolite pairs.	[ ]
Accuracy	Mass calibration verified against lock mass.	Absolute error $\lvert\delta_{\text{ppm}}\rvert \leq 5.0$ ppm.	[ ]
Strategy	Acquisition mode matches statistical model.	DDA modeled with MNAR/MAR; DIA deconvoluted.	[ ]
Suppression	Stable isotope labeled standards tracked.	Internal standard CV $\leq 15\%$ across all runs.	[ ]

2.8 Chapter Exercises

Exercise 2.1 (Charge State and Neutral Mass Reconstruction). An unknown tryptic peptide is detected on an LC-MS system. The monoisotopic peak appears at $m/z = 624.8348$, and the first ^{13}C isotopic peak appears at $m/z = 625.1693$.

1. Determine the peak spacing $\Delta(m/z)$ and calculate the integer charge state z.

2. Reconstruct the monoisotopic neutral mass (M_{neutral}) in Daltons.

Exercise 2.2 (Resolving Power for Isobaric Metabolites). Two small molecules have identical nominal mass (132 Da) but distinct elemental compositions: Metabolite 1 (132.0773 Da) and Metabolite 2 (132.1025 Da).

1. Calculate the exact mass difference Δm.

2. Compute the minimum mass resolving power R required for baseline separation and state which mass analyzers satisfy this requirement.

Exercise 2.3 (Stochastic Missingness Mechanisms in DDA). Explain why Data-Dependent Acquisition (DDA) produces stochastic missing values across replicate runs of the same biological sample. Categorize this missingness under Rubin's missing data taxonomy (MCAR, MAR, MNAR).

Exercise 2.4 (Matrix Suppression Correction via Isotope Dilution). A targeted LC-MS assay suffers 60% electrospray ion suppression ($S(t) = 0.40$) during the elution window of an endogenous analyte.

1. Write the mathematical response equations for the endogenous analyte and a co-eluting stable isotope-labeled (SIL) internal standard.

2. Prove that the area ratio $I_{\text{analyte}}/I_{\text{SIL}}$ is invariant to the suppression factor.

2.9 Worked Solutions to Chapter Exercises

2.9.1 Solution to Exercise 2.1: Charge State and Neutral Mass Reconstruction

Part 1: Charge State Determination The observed isotopic peak spacing is:

$$\Delta(m/z) = 625.1693 - 624.8348 = 0.3345 \text{ Da} \tag{2.10}$$

The integer charge state is:

$$z = \text{round}\left(\frac{1.003355}{0.3345}\right) = \text{round}(2.9996) = 3 \tag{2.11}$$

Part 2: Neutral Mass Calculation Subtracting 3 protons ($m_{\text{proton}} = 1.007276$ Da):

$$M_{\text{neutral}} = 3 \times 624.8348 - 3 \times 1.007276 = 1874.5044 - 3.0218 = 1871.4826 \text{ Da} \tag{2.12}$$

2.9.2 Solution to Exercise 2.2: Resolving Power for Isobaric Metabolites

Part 1: Mass Difference

$$\Delta m = 132.1025 - 132.0773 = 0.0252 \text{ Da} = 25.2 \text{ mDa} \tag{2.13}$$

Part 2: Resolving Power Requirement

$$R \geq \frac{132.0773}{0.0252} \approx 5,241 \tag{2.14}$$

A unit-mass quadrupole ($R \approx 1,000$) fails to resolve the doublet, merging them into a single envelope. A Time-of-Flight ($R \geq 30,000$) or Orbitrap ($R \geq 60,000$) resolves the doublet completely.

2.9.3 Solution to Exercise 2.3: Stochastic Missingness Mechanisms in DDA

In DDA, the mass spectrometer dynamically selects only the top N most abundant precursor ions detected in each survey scan. Because low-abundance peptides fluctuate near the selection threshold, small run-to-run variations in ionization or chromatographic apex timing determine whether a peptide is selected or excluded. Under Rubin's taxonomy:

- Peptides below the instrument detection limit exhibit Missing Not At Random (MNAR) censoring.

- Peptides with sufficient abundance that fluctuate across the Top-N boundary exhibit Missing at Random (MAR) stochasticity conditioned on instantaneous MS1 intensity.

2.9.4 Solution to Exercise 2.4: Matrix Suppression Correction via Isotope Dilution

Part 1: Response Equations Let k denote the instrument response factor. Under 60% suppression ($S = 0.40$):

$$I_{\text{analyte}} = 0.40 \cdot k \cdot C_{\text{analyte}} \tag{2.15}$$

$$I_{\text{SIL}} = 0.40 \cdot k \cdot C_{\text{SIL}} \tag{2.16}$$

Part 2: Invariance Proof Computing the signal ratio:

$$\frac{I_{\text{analyte}}}{I_{\text{SIL}}} = \frac{0.40 \cdot k \cdot C_{\text{analyte}}}{0.40 \cdot k \cdot C_{\text{SIL}}} = \frac{C_{\text{analyte}}}{C_{\text{SIL}}} \tag{2.17}$$

The suppression coefficient 0.40 and instrument factor k cancel algebraically, preserving linear calibration accuracy.

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